that it is the square–root of the mean of the squared deviation from the arithmetic mean. It provides accurate result. Square of standard deviation is called Variance.

Definition:

It is defined as the positive square-root of the arithmetic mean of the square of the deviations of the given observation from their arithmetic mean.

The standard deviation of the population is denoted by the Greek letter **σ (sigma)** The standard deviation of sample is denoted as 's'.

Calculation of Standard deviation-Individual Series:

There are two methods of calculating Standard deviation in an individual series.

a) Deviations taken from Actual mean

b) Deviation taken from Assumed mean

$$\text{Standard Deviation} = \sqrt{\left(\frac{\sum(x-\bar{x})^2}{n}\right)}$$

Where, $\bar{x}$ = mean value of distribution

n = number of observations.

Steps:

1. Find out the actual mean of given data ($\bar{X}$)
2. Find out the deviation of each value from the mean ($x = X - \bar{X}$)
3. Square the deviations and take the total of squared deviations $\sum x^2$
4. Divided the total $\sum x^2$ by the number of observation $\left(\frac{\sum x^2}{n}\right)$
5. The square root of $\left(\frac{\sum x^2}{n}\right)$ is standard deviation.

$$\sigma = \sqrt{\left(\frac{\sum x^2}{n}\right)} \quad \text{or} \quad \sqrt{\left(\frac{\sum(x-\bar{x})^2}{n}\right)}$$

$$\frac{\sum x^2}{n} = \text{Variance} = \frac{\sum(x-\bar{x})^2}{n}$$

When the sample size is less than 30,

$$\text{variance} = \frac{\sum(x-\bar{x})^2}{n-1};$$

When n = number of observations.

Example 1: Calculate the standard deviation from the following data by **Actual Mean Method**: 25, 15, 23, 42, 27, 25, 23, 25, and 20.

Solution: Deviations from actual mean.

Sl. No	Values (X)	$X - \bar{X}$	$(X - \bar{X})^2$
1	25	25-25=0	0
2	15	15-25=10	100
3	23	23-25= -2	4
4	42	42-25=17	289
5	27	27-25=2	4
6	25	25-25=0	0
7	23	23-25=-2	4
8	25	25-25=0	0
9	20	20-25=-5	25
N=9	225	0	426

$$\bar{X} = \frac{225}{9} = 25$$

$$\sigma = \sqrt{\left(\frac{\sum(x-\bar{x})^2}{n}\right)} = \sqrt{\frac{426}{9}} = \sqrt{47.33}$$

σ = 6.88 Answer

Example 2: Calculate the standard deviation for the following data by assumed mean method: 43, 48, 65, 57, 31, 60, 37, 48, 78, 59

12-Economics-Chapter_12.indd 264 23-12-2022 14:48:30